Kush Rishi

GNSS Analyst at Xona. BSc in Geomatics Engineering, University of Calgary.

Montréal · kushrishi04@gmail.com · kushrishi.com · github.com/Kushrishi

Experience

GNSS Analyst · Xona

Montréal · June 2026–present

Develop Python and Linux workflows for GNSS collection, processing, monitoring and validation. Build data-quality checks, engineering metrics and dashboards; support receiver integration and troubleshooting.

Research Assistant · University of Calgary

December 2025–April 2026

Processed LiDAR and photogrammetric point clouds in CloudCompare, calculated spatial quality metrics and prepared documented datasets and visualizations for faculty research.

Geomatics Team Member · Relectric Car Team

September 2023–April 2026

Supported electric-vehicle conversion with spatial measurements and geometry models. Worked with GNSS, IMU and LiDAR data and prepared technical outputs for other subteams.

Survey Assistant · McElhanney

May–August 2024

Supported control, layout and as-built surveying for the Valley Line West LRT extension using GNSS receivers, total stations and digital levels.

Software Application Developer · Hycroft Chiropractic & Massage

September 2021–June 2022 · Freelance

Built and maintained the Stretch2Go application in Dart, including booking, exercise plans, journaling and content updates. Refined user flows through testing and feedback.

Education

University of Calgary

BSc Geomatics Engineering, With Distinction · 2022–2026

President, Geomatics Engineering Students' Society; Engineering Students' Society Board of Directors, 2025–2026.

Selected projects

Model Regression Forensics

Built a prediction importer, release-policy checks and local reports. In two constructed Banking77 benchmarks, model-based diagnostics added no consistent top-1 benefit over simple baselines. A separate example demonstrates ambiguous repairs; general causal identification is not established.

Autonomy Simulation Lab

Built a browser planning/localization simulator and a separate C++/Python recording-replay tool with CPU inference and configuration comparison. A crop-aware optimization reduced median whole-process time by 8.97% across three matched 108-frame pairs with exact recorded output parity on one Linux host. Measurements apply to that host, model and workload.

Interactive demo

TrueMargin

Evaluated registration variability against known spatial error in 30 controlled cases across 10 anatomies, followed by separate calibration and evaluation cohorts. Median anatomy-level rank correlation was 0.6841; six case-level associations were negative. Bounds were conservative, and no advantage over inverse-consistency error was established. Independent external validation was not completed.

Technical focus

Python, C++, PyTorch, Linux, Git, CI, data pipelines, numerical evaluation, GNSS/PNT, localization, state estimation and image registration.